

User Requirements Specification For The Lost & Found Item Tracker Serving Group 8

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1. Introduction

1.1 Objectives

This is the User Requirements Specification for the Lost & Found Item Tracker system. These requirements will ensure that the system will correctly and reliably perform its intended functionality.

The objective of the Lost & Found Item Tracker system is to create a web-based platform for reporting, tracking, and claiming lost items. The ability to connect users to their lost items must be a simple and fast process through an organized and searchable system.

1.2 Scope

This User Requirements Specification applies to the Lost & Found Item Tracker system. The URS will address the functions the system must be able to perform successfully to perform the primary functionality. This document also addresses the regulatory requirements for a system like the Lost & Found Item Tracker.

1.3 Assumptions

The application will be running on a properly functioning computer or mobile device, with a consistent network connection

1.4 Exclusions

This validation applies to the Lost & Found Item Tracker system and not to the various web browsers that may be used to access the application.

2. Program Requirements

2.1 Core Functional Requirements

- **PR-01:** The system shall allow a user to submit a found item through a form that collects details, including the item's title, description, location, date found, category, and an optional photo.
- **PR-02:** The system shall display all submitted items in a centralized, searchable list accessible to any authenticated user.
- **PR-03:** The system shall allow users to filter and search for items based on the following:
 - Category (default: *"all"*)
 - Location (default: *unspecified*)
 - Date found (default: *newest-first*)
 - Whether items have been marked as "claimed" (default: *disabled*)

- **PR-04:** The system shall allow an item's status to be updated to "claimed" once it has been returned to the owner, which will, in turn, remove the post from "default" public search results.

2.2 Usability Requirements

- **PR-05:** The item submission and search workflows shall be designed to require no more than three user actions from the main screen.
- **PR-06:** The system shall be responsive and optimized for mobile browsers, ensuring accessibility across devices.

2.3 Security Requirements

- **PR-07:** The system shall verify user authenticity through a valid login credential.
- **PR-08:** Sensitive information, such as user contact details and message contents, shall only be viewable by involved parties.
- **PR-09:** The system shall provide a means to expedite conflict reporting and/or resolution, wherein a "conflict" could include:
 - Incorrect or intentionally false claims of ownership
 - Deceitful "found" claims
 - User harassment

3. Data Requirements

3.1 Item Data

- **DR-01:** Each item record shall have a unique, system-generated Item ID that identifies it within the database.
- **DR-02:** Each record shall store the Item Title, Description, Category, Location, and Date Found or Date Lost.
- **DR-03:** Each record shall include a Status field (e.g., "Found," "Lost," "Claimed," "Archived") that reflects its current state.
- **DR-04:** The system shall store an optional photo for each item.
- **DR-05:** Each item shall store a Reporter ID referencing the user who created the listing to ensure accountability.
- **DR-06:** The system shall record timestamps for creation, last modification, and claim completion.
- **DR-07:** Claimed or inactive items shall remain accessible in an archived state for administrative review and audit purposes.

3.2 User Data

- **DR-08:** Each user shall have a unique User ID and authenticated University Email Address.
- **DR-09:** The system shall store the user's Display Name.
- **DR-10:** Each user shall have an assigned Role (e.g., Standard User, Administrator, Moderator) that governs access permissions.

- **DR-11:** The system shall track User Activity Logs for submission, modification, and claim events to support traceability.
- **DR-12:** Personally identifiable information, including names and emails, shall only be accessible to administrators and not displayed publicly without consent.

3.3 Communication Data

- **DR-13:** Each message shall have a unique Message ID and be associated with a Conversation ID between two users.
- **DR-14:** Each message shall include Sender ID, Receiver ID, Timestamp, and Message Content.
- **DR-15:** These records shall only be accessible to the involved users and administrators for moderation purposes
- **DR-16:** Notification records shall be stored when a user receives a message, claim update, or new item alert.

3.4 Data Integrity

- **DR-17:** All user-submitted input shall undergo server-side validation to ensure required fields are complete and contain valid data types.
- **DR-18:** The database shall enforce referential integrity between users, items, and messages
- **DR-19:** User authentication tokens or password hashes shall be stored using industry-standard encryption
- **DR-20:** The system shall maintain a consistent timestamp standard (e.g., UTC) across all data entries to ensure chronological accuracy.

4. Life Cycle Requirements

4.1 Updates and Scalability

- **LCR-01:** The system's architecture and database schema shall be designed to accommodate additional data fields and features without major redesign

4.2 Decommissioning

- **LCR-02:** If the system should be replaced or discontinued, a structured decommissioning process shall ensure safe removal of user data, following data protection standards.

5. Regulatory Requirements

In the event that the Lost & Found Item Tracker system is implemented beyond the student project and put into use by the student body at UW-Milwaukee, the following regulatory requirements should be implemented.

5.1 Privacy & Data Protection

In order for the system to comply with the FTC Act §5 against “unfair or deceptive practices”, the system would have to post a Privacy Policy that explains what personal data is collected by the system, why this data is collected, who is able to access the data, and how users can delete their data or close accounts. The data must also be stored securely with encrypted login information and limited access to user information.

5.2 Terms of Service (ToS) / User Agreement

In order to better protect the users of the platform, there should be a ToS Agreement created to define user responsibilities, ban illegal content or misuse of the system, and outline how to suspend or delete accounts that violate the terms listed. The ToS will be able to be upheld via a content moderation policy for uploaded images and descriptions to be reviewed if they violate the ToS or are images not allowed on the system.

5.3 Accessibility Requirements

The system should try to follow the Web Content Accessibility Guidelines version 2.1 (WCAG 2.1) for accessibility requirements. These include that alt text is added to the images of lost items, and the system uses sufficient color contrast (minimum 4.5:1 for text). The system should have form fields and buttons labeled clearly, and the text should resize well with changes to the window size and on mobile platforms.

6. Workflow

The Lost & Found Item Tracker will follow a straightforward workflow emphasizing ease of use and traceability. The main user interactions are described below.

6.1 Reporting a Found Item

1. The user navigates to the “Submit Found Item” page.
2. A form to collect item details, including a title, description, location, date found, category, and an optional photo, is displayed to the user.
3. Upon submission, the form's data is validated and stored in the database along with a generated item ID.
4. After a submission is completed, a confirmation message is displayed to the user, and the item is added to the searchable list.

6.2 Searching for Lost Items

1. The user navigates to the “Find Lost Items ” page.

2. A search interface listing “found items” is displayed to the user with filters for category, location, and date.
3. Users can select an item from the list to view more details, including the item description, image, and the means to contact and claim the item.
4. If an item's ownership is verified, the item's status is updated and marked as “claimed,” archiving the item and removing it from the public search list.

6.3 Claiming Items

1. When a potential match between a user and a found (lost) item in the system is found, the user can reach out to the finder using the internal messaging system.
2. The user can then send a claim request via the item’s page.
3. Once a user is verified as the owner of a found (lost) item, the item’s status is changed to “claimed,” archiving the item and removing it from the public search list.

6.4 Admin Oversight

- Admins and moderators can manage user permissions, review flagged content, examine archived data, and see system logs for submissions and claims for traceability and user disputes.

7. Stretch Goals

The following features are potential features that may be added to the Lost & Found Item Tracker in the future to further improve the user experience.

7.1 Lost-Item Alerts and Notifications for Potential Matches

Implementation of a background system that analyzes and matches users with potential lost (found) items based on criteria provided by the user. When similarities are detected between the user's query criteria and the description of a newly submitted item, the user will be notified within the app of a potential match.

7.2 Refined UI/UX for Desktop Experience

Implementation of a UI/UX designed for a desktop browser.

7.3 Campus Collaboration for Item Drop-Off and Retrieval

Partner with UWM’s lost and finds along with campus security to integrate the designated campus lost and found locations as drop-off and pickup zones, allowing for scheduled pickups and further verification of ownership of claimed items.

7.4 Image Matching to Suggest Similar Items

Implement AI image recognition that searches the images in the list of lost (found) items for items that are similar to a user input image to reduce manual search efforts.

8. Conclusion

The Lost & Found Item Tracker program proposed in this User Requirement Specification document is designed to provide a streamlined and user-friendly means of reporting, locating, and reclaiming lost items from within the UWM community in a stress-free and intuitive fashion. This document outlines and defines the Lost & Found Item Tracker program's necessary functional, data, and regulatory requirements to carry out the system's scalability, security, and user accessibility.

Our design specifications promote efficiency and accountability through secure user authentication as well as the implementation of administrative oversight to ensure that user disputes are properly handled. The planned stretch goals for possible future implementation, including notifications, improved desktop UI/UX, campus collaboration, and AI image matching, were conceived with the goal of further improving the efficiency with which users can find their lost items in order to create a better user experience.

Overall, this document acts as the foundation for the design, implementation, sustainability, and successful deployment of the Lost & Found Item Tracker program.

9. Citations

Federal Trade Commission act section 5: Unfair or deceptive acts or practices. (n.d.).
<https://www.federalreserve.gov/boarddocs/supmanual/cch/200806/ftca.pdf>

Web content accessibility guidelines (WCAG) 2.1. W3C. (n.d.).
<https://www.w3.org/TR/WCAG21/>